

Voice Based Concept Understanding Analyser

Project Description:

Voice Based Concept Understanding Analyser is an AI-powered educational application that analyzes a learner's spoken explanation of a concept and evaluates how well it matches the expected topic. The application converts speech into text using OpenAI Whisper, processes the transcript using Sentence Transformers to calculate semantic similarity with the reference concept, and presents the understanding score to the user. The system also generates a detailed PDF report containing the transcript, similarity score, and concept analysis. The application is developed using Python and Streamlit with supporting libraries such as Librosa, SoundFile, Matplotlib, and ReportLab, providing an interactive and user-friendly platform for concept assessment.

Scenarios

Scenario 1: A student uploads an audio recording explaining **Machine Learning**. The application converts the speech into text, compares it with the reference concept, and displays the similarity score along with feedback.

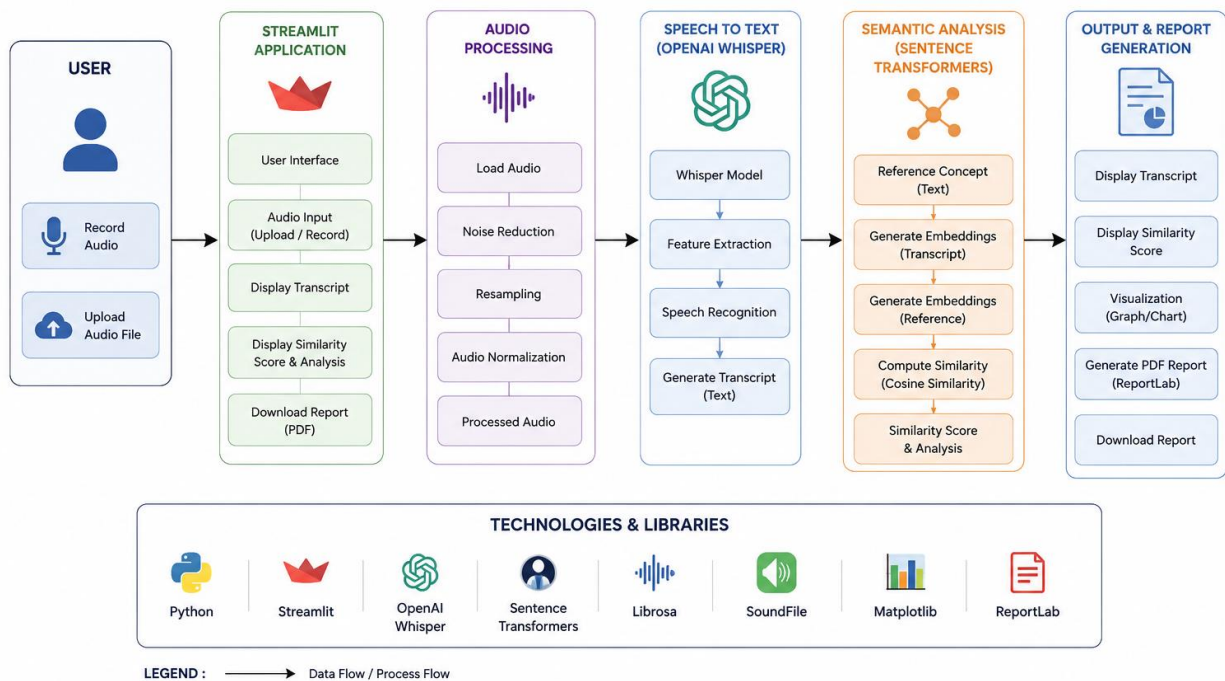
Scenario 2: A faculty member evaluates students' spoken explanations during a viva examination. The system analyzes each recording and generates an assessment report for every student.

Scenario 3: An online learner practices explaining technical concepts. The application provides immediate feedback on concept understanding and allows the learner to improve their explanations.

Technical Architecture:

TECHNICAL ARCHITECTURE

VOICE BASED CONCEPT UNDERSTANDING ANALYSER



Description: The application follows a modular architecture consisting of a Streamlit-based user interface, a Python backend, an audio preprocessing module, a speech-to-text module using OpenAI Whisper, a semantic similarity module based on Sentence Transformers, and a report generation module. Audio files uploaded by the user are preprocessed using Librosa and SoundFile before transcription. The generated transcript is analyzed for semantic similarity against reference content. The similarity score, transcript, and analysis are presented to the user through the Streamlit interface, while ReportLab and Matplotlib generate downloadable PDF reports.

Pre-requisites:

- Python 3.10 or above
- Streamlit
- OpenAI Whisper
- Sentence Transformers
- Librosa
- SoundFile
- Matplotlib
- ReportLab
- Git
- Visual Studio Code

Project Workflow:

Milestone 1 – Project Planning and Environment Setup

Activities:

- Install Python and required libraries.
- Create a virtual environment.
- Install Streamlit and AI libraries.
- Design overall project architecture.
- Organize project folder structure.

Milestone 2 – Audio Processing Module

Activities

- Develop audio upload interface.
- Implement audio preprocessing.
- Handle different audio formats.
- Improve audio quality before transcription.

Milestone 3 – Speech-to-Text Module

Activities

- Integrate OpenAI Whisper.
- Convert speech into text.
- Display transcription.
- Handle transcription errors.

Milestone 4 – Concept Understanding Module

Activities

- Integrate Sentence Transformers.
- Generate embeddings.
- Compare transcript with reference concept.
- Calculate semantic similarity score.

Milestone 5 – Report Generation

Activities

- Generate similarity report.
- Create graphical analysis.
- Export report as PDF.
- Display results in Streamlit.

Milestone 6 – Testing and Deployment

Activities

- Perform unit testing.
- Verify similarity calculations.
- Test report generation.
- Deploy application locally.

App Development

Overview

Description

The Voice Based Concept Understanding Analyser is developed using Python and Streamlit to provide an interactive platform for evaluating a learner's conceptual understanding through speech. The application accepts an audio explanation from the user, converts the speech into text using OpenAI Whisper, analyzes semantic similarity using Sentence Transformers, evaluates fluency metrics, generates graphical visualizations, and creates a downloadable PDF report.

Activity 1 – User Interface Development

Description

A user-friendly interface was designed using Streamlit. The interface provides project information, allows users to select a concept, supports multiple audio formats, and enables audio file uploads for analysis.

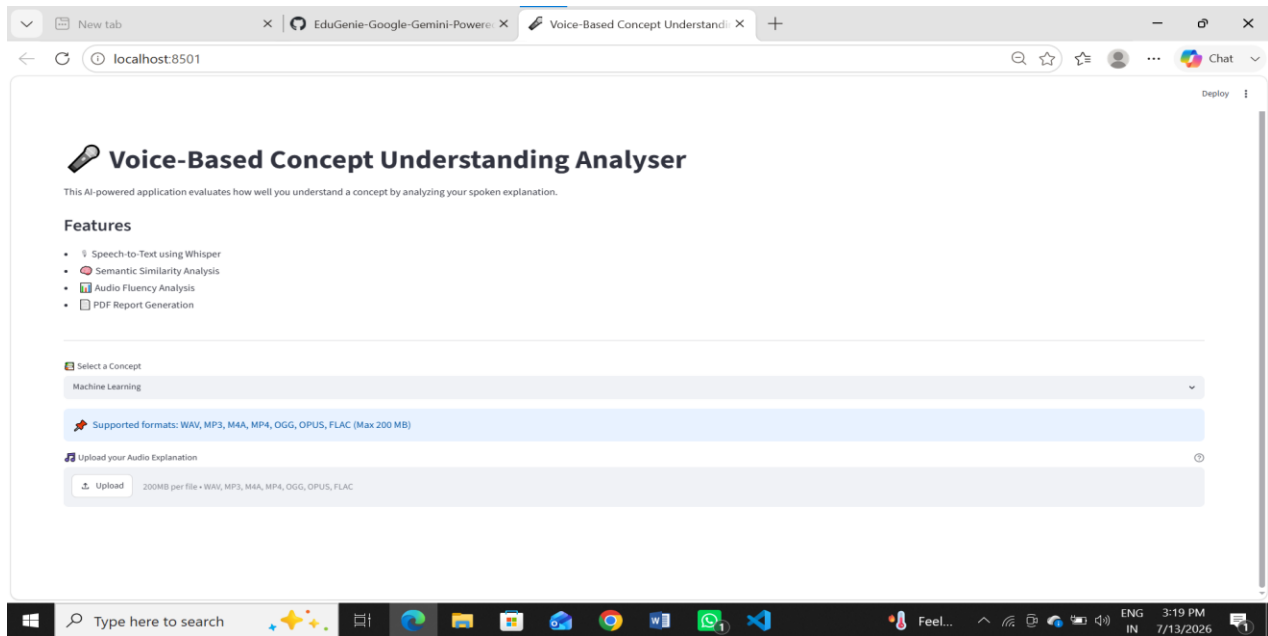


Figure 1. Home page of the Voice Based Concept Understanding Analyser

Activity 2 – Audio Upload

Description

Users can upload audio explanations in supported formats such as WAV, MP3, MP4, M4A, OGG, OPUS, and FLAC. Once uploaded successfully, the application stores the audio and prepares it for processing.

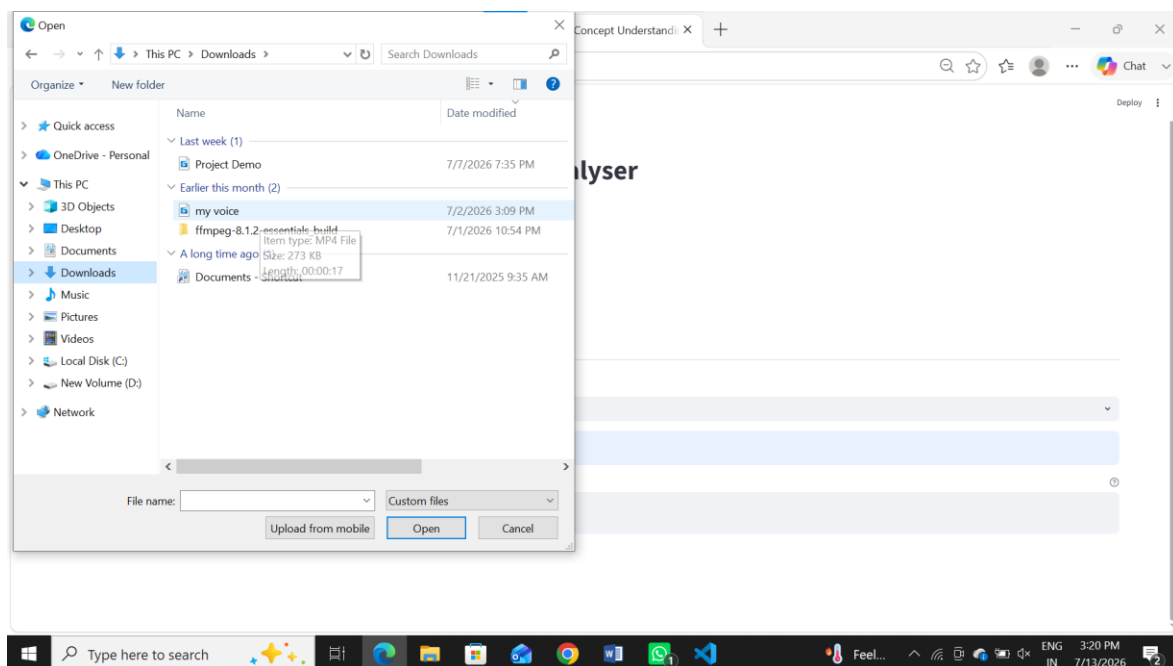


Figure 2. Audio upload and preprocessing interface

Activity 3 – Speech-to-Text and Concept Analysis

Description

The uploaded speech is transcribed using OpenAI Whisper. The generated transcript is compared with the selected reference concept using Sentence Transformers. The application then computes the semantic similarity score and evaluates the learner's understanding level along with fluency metrics.

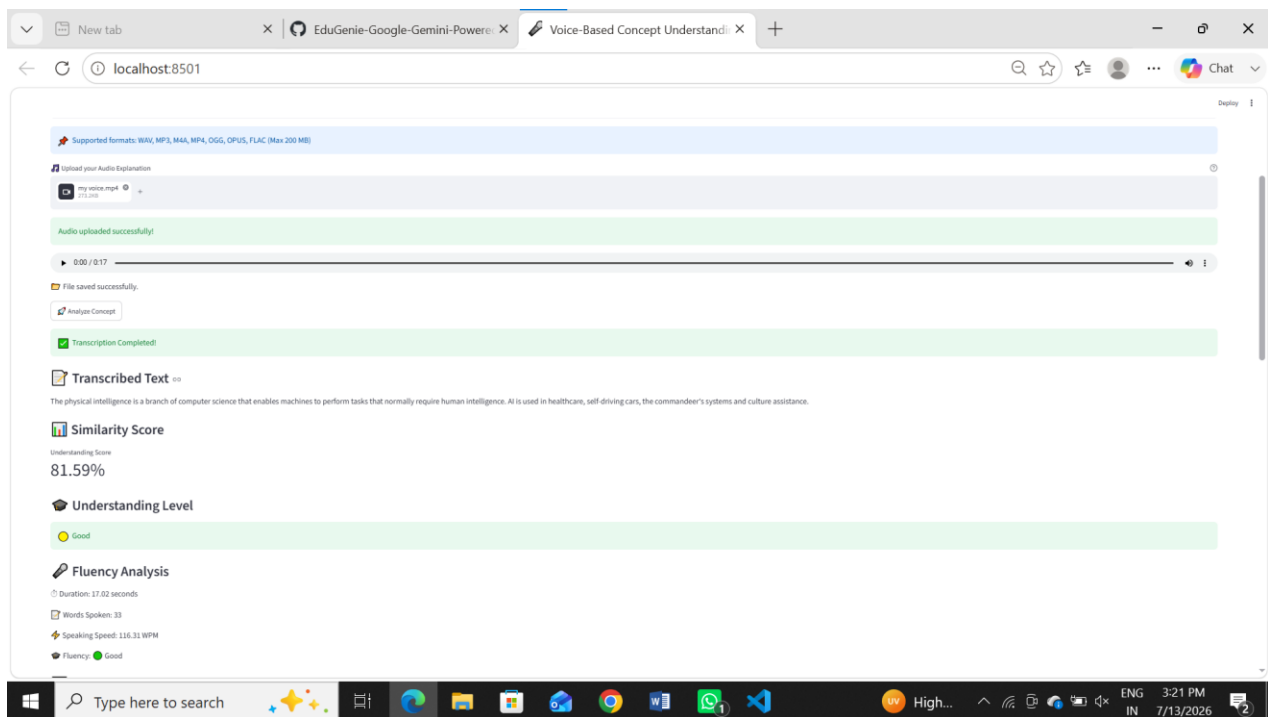


Figure 3. Speech transcription, semantic similarity score, and fluency analysis

Activity 4 – Audio Waveform Analysis

Description

The system visualizes the uploaded audio waveform to represent amplitude variation over time. This visualization assists in analyzing voice characteristics before generating the final report.

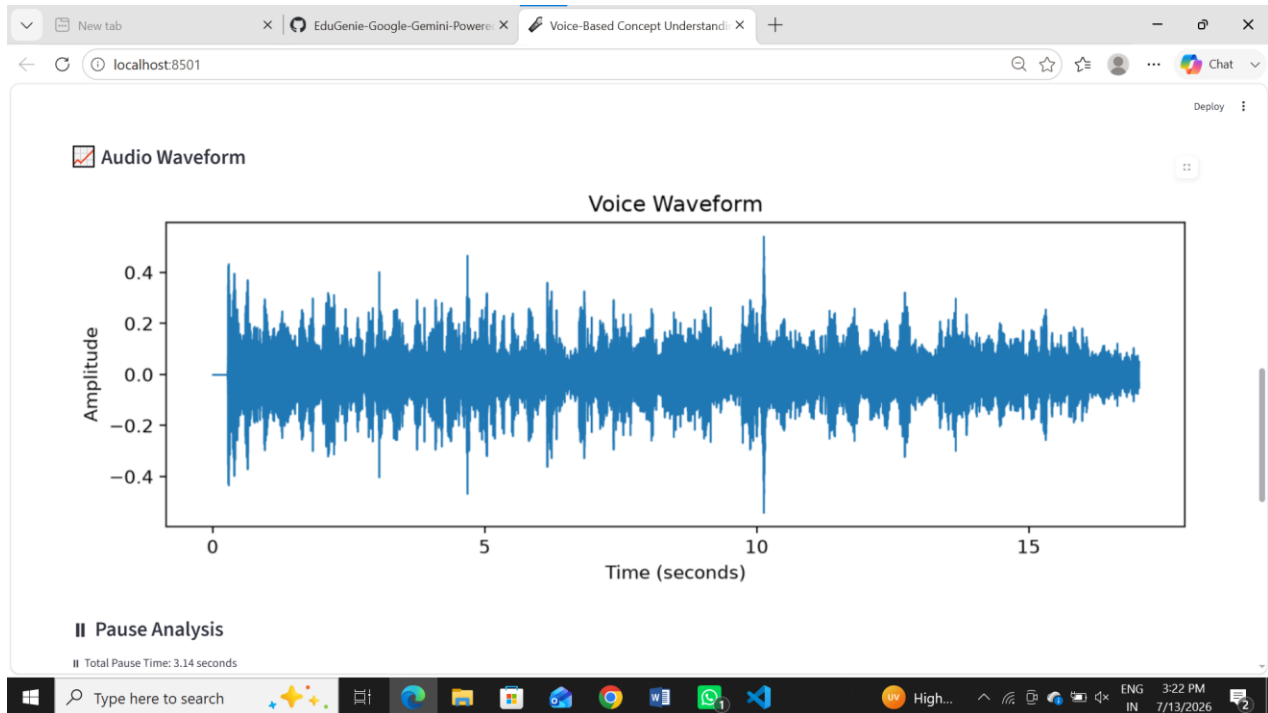


Figure 4. Audio waveform visualization

Activity 5 – Performance Summary

Description

After completing the analysis, the application generates an AI-based performance summary that includes pause analysis, RMS energy analysis, filler word detection, and an overall assessment of the user's spoken explanation.

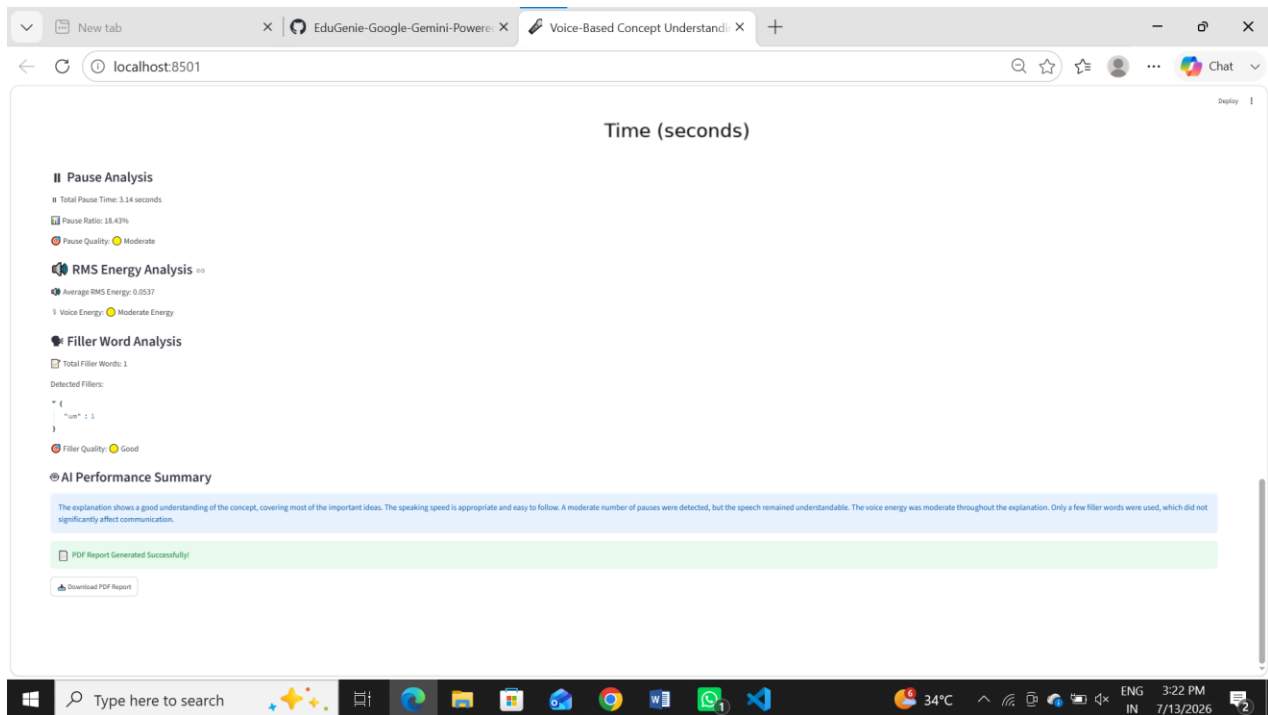


Figure 5. AI-generated performance summary and speech quality analysis

Activity 6 – PDF Report Generation

Description

The final analysis is compiled into a downloadable PDF report. The report contains the transcribed text, semantic similarity score, fluency metrics, pause analysis, energy analysis, waveform visualization, and overall performance summary.

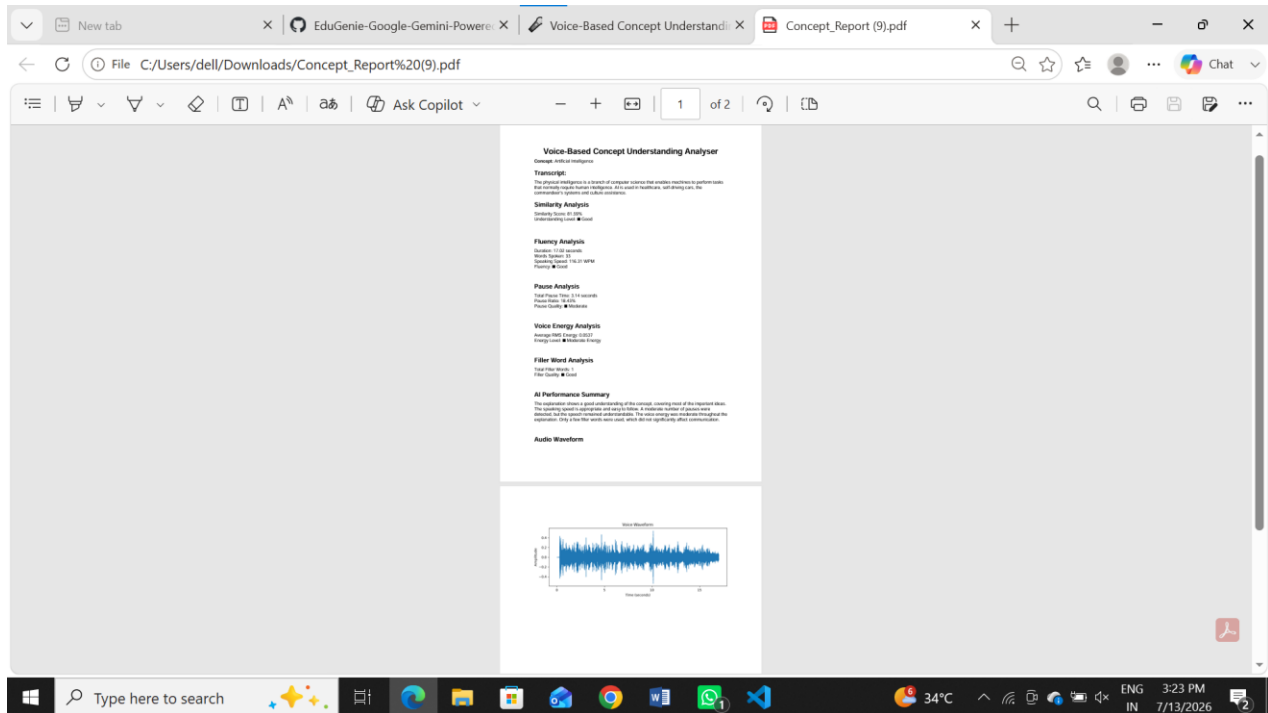


Figure 6. Generated concept analysis PDF report

Deployment

Description

The application is executed locally using Streamlit. After installing the required dependencies, the application can be launched using the Streamlit server and accessed through a web browser.

Testing

Test Case	Expected Result	Status
Launch Application	Home page loads successfully	Pass
Upload Audio File	Audio uploaded without errors	Pass
Speech-to-Text	Transcript generated correctly	Pass
Semantic Similarity	Understanding score calculated	Pass
Fluency Analysis	Speech metrics displayed	Pass
PDF Generation	PDF report generated successfully	Pass

Conclusion

The Voice Based Concept Understanding Analyser successfully evaluates a learner's spoken explanation using Artificial Intelligence techniques. By integrating OpenAI Whisper for speech-to-text conversion, Sentence Transformers for semantic similarity analysis, and Streamlit for the user interface, the application provides accurate concept assessment along with fluency evaluation and automated PDF report generation. The project demonstrates the practical application of AI in educational assessment and offers an effective tool for students and educators to evaluate conceptual understanding through spoken communication.